

Annex 120G

(normative)

Chip-to-module 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2M), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2M), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2M)

120G.1 Overview

This annex defines the functional and electrical characteristics for the optional chip-to-module (C2M) 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2M), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2M), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2M). Figure 120G–1 shows the relationship of the 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2M interfaces to the ISO/IEC Open System Interconnection (OSI) reference model. The C2M interface provides electrical characteristics and associated compliance points, which can optionally be used when designing systems with pluggable module interfaces.

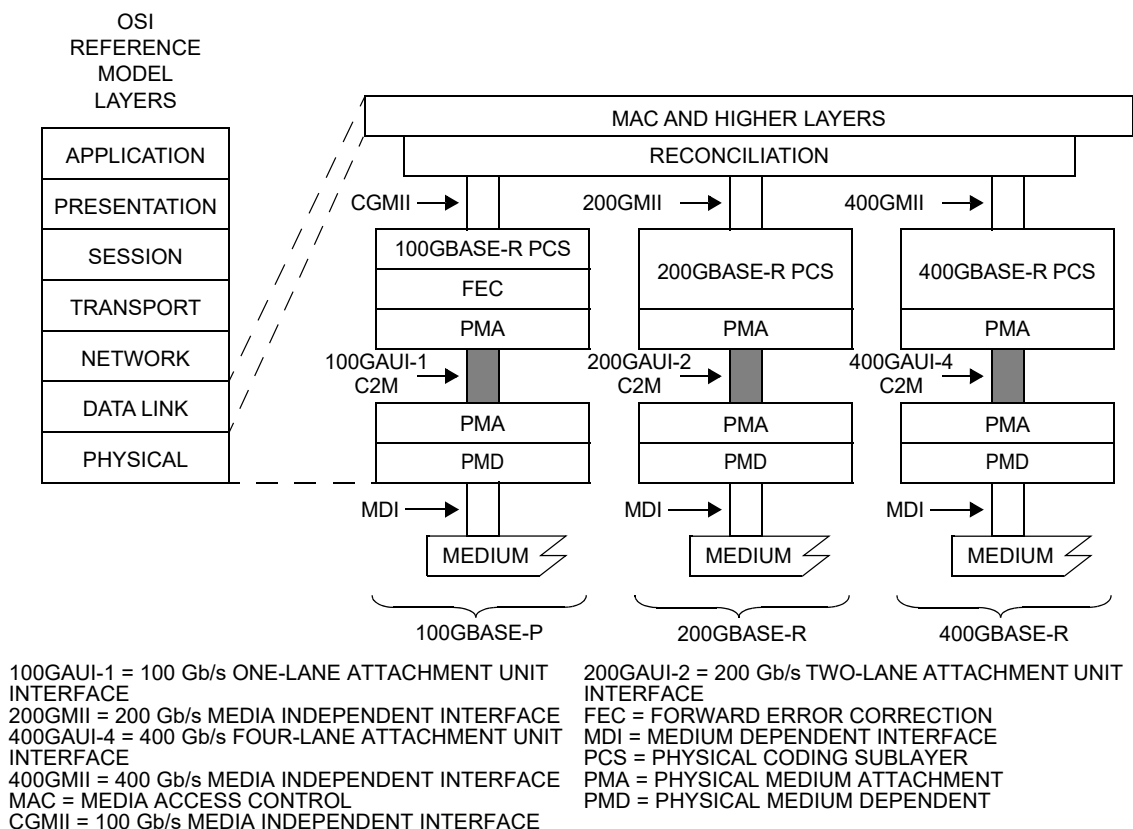
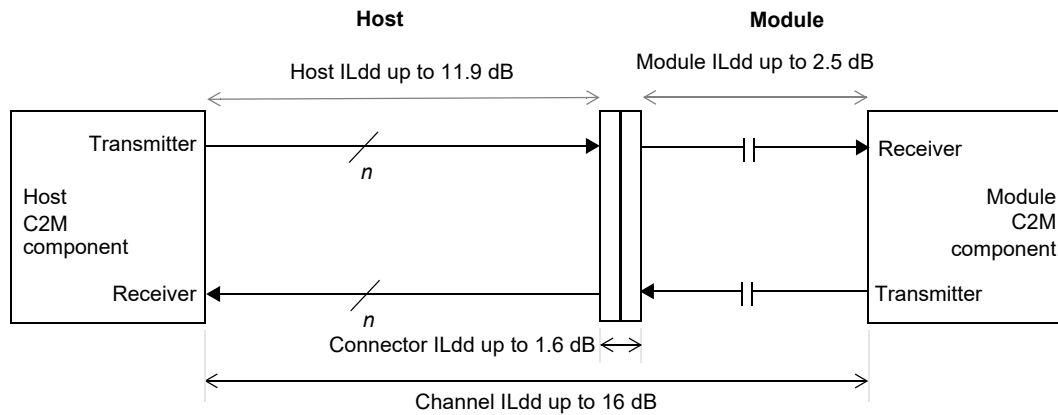


Figure 120G–1—Example 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2M relationship to the ISO/IEC Open System Interconnection (OSI) reference model and the IEEE 802.3 Ethernet model

The sublayers including PCS and FEC (for 100GBASE-P PHYs) for each PHY that can optionally include a 100GAUI-1 C2M, 200GAUI-2 C2M, or 400GAUI-4 C2M are summarized in the tables in 80.1.5 and 116.1.4 and are specified in the corresponding PMD clause. The positioning of the 100GAUI-1 C2M, 200GAUI-2 C2M, or 400GAUI-4 C2M relative to other sublayers is shown in 135.1 and 120.1 with further examples in Annex 135A and Annex 120A.

The C2M link is described in terms of a host C2M component, a C2M channel with associated insertion loss (ILdd), and a module C2M component. The host C2M component and module C2M component have connected ground references. Figure 120G–2 depicts a typical C2M application and summarizes the ILdd budget associated with the C2M application. The ILdd budget reflects the highest ILdd intended to be supported and is characterized by Equation (120G–5) and illustrated in Figure 120G–12. The C2M interface comprises independent data paths in each direction. Each 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2M data path contains one, two, or four differential lanes, respectively, using PAM4 signaling, where the highest differential level corresponds to the symbol three and the lowest level corresponds to the symbol zero. Each lane is AC-coupled within the module.



NOTE—The number of lanes n is equal to 1 for 100GAUI-1, 2 for 200GAUI-2, and 4 for 400GAUI-4.

Figure 120G–2—100GAUI-1, 200GAUI-2, and 400GAUI-4 C2M insertion loss budget at 26.56 GHz

The nominal signaling rate for each lane is 53.125 GBd.

120G.1.1 Bit error ratio

The bit error ratio (BER) when processed by the PMA according to Clause 135 for 100GAUI-1 C2M or Clause 120 for 200GAUI-2 or 400GAUI-4 C2M shall be less than 10^{-5} .

120G.2 Compliance point definitions

The electrical characteristics for the 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2M interfaces are defined at compliance points for the host and module. Reference test fixtures, called compliance boards, are used to access the electrical specification parameters.

Figure 120G–3 depicts the location of compliance points when measuring 100GAUI-1, 200GAUI-2, or 400GAUI-4 C2M host compliance. The output of the Host Compliance Board (HCB) is used to verify the host electrical output signal at TP1a. The input of the HCB at TP4a is used to verify the host input compliance.

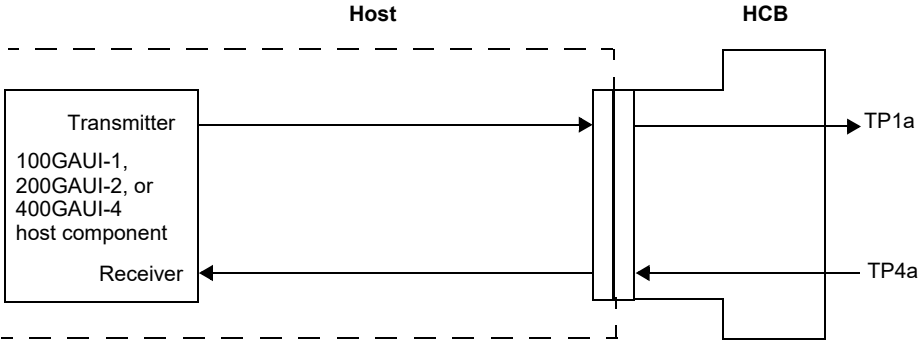


Figure 120G–3—Host compliance points

Figure 120G–4 depicts the location of compliance points when measuring 100GAUI-1, 200GAUI-2, or 400GAUI-4 C2M module compliance. The output of the Module Compliance Board (MCB) is used to verify the module electrical output signal at TP4. The input of the MCB at TP1 is used to verify the module input compliance.

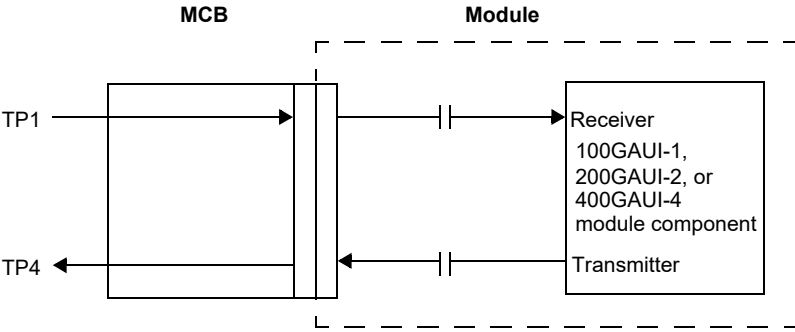


Figure 120G–4—Module compliance points

Additional details on the requirements for the MCB and HCB are given in 120G.5.4.

120G.3 Electrical characteristics

120G.3.1 Host output characteristics

The host output shall meet the specifications given in Table 120G–1.

A test system with a fourth-order Bessel-Thomson low-pass response with 40 GHz 3 dB bandwidth is to be used for all output signal measurements, unless otherwise specified.

Table 120G–1—Host output characteristics at TP1a

Parameter	Reference	Value	Units
Signaling rate, each lane (range)		53.125 ± 50 ppm ^a	GBd
DC common-mode output voltage (max)	120G.5.1	2.8	V
DC common-mode output voltage (min)	120G.5.1	−0.3	V
Single-ended output voltage (max)	120G.5.1	3.3	V
Single-ended output voltage (min)	120G.5.1	−0.4	V
Peak-to-peak AC common-mode voltage (max) Low-frequency, $V_{CM_{LF}}$ Full-band, $V_{CM_{FB}}$	120G.5.1	32 80	mV
Differential peak-to-peak output voltage (max) Transmitter disabled Transmitter enabled	120G.5.1	35 750	mV
Steady-state voltage, v_f (max)	120G.5.3	375	mV
Eye height (min)	120G.3.1.5	10	mV
Vertical eye closure, VEC (max)	120G.3.1.5	12	dB
Common-mode to differential-mode return loss, RL_{dc} (min)	120G.3.1.1	Equation (120G–1)	dB
Effective return loss, ERL (min)	120G.3.1.2	7.3	dB
Differential termination mismatch (max)	120G.3.1.3	10	%
Transition time (min) Host is requesting short mode ^b Host is requesting long mode ^b	120G.3.1.4	10 15	ps ps

^a For a PMA in the same package as the PCS sublayer. In other cases, the signaling rate is derived from the signaling rate presented to the PMA input lanes (see Figure 135–3 and Figure 120–3) by the adjacent PMA or FEC sublayers.

^b Short and long modes are defined in 120G.3.2.1.

120G.3.1.1 Host output common-mode to differential-mode return loss

The host output common-mode to differential-mode return loss shall meet Equation (120G–1) as illustrated in Figure 120G–5.

$$RL_{dc}(f) \geq \begin{cases} 25 - 22(f/53.125) & 0.05 \leq f \leq 26.56 \\ 19 - 10(f/53.125) & 26.56 < f \leq 50 \end{cases} \quad (120G-1)$$

where

RL_{dc} is the common-mode to differential-mode return loss in dB
 f is the frequency in GHz

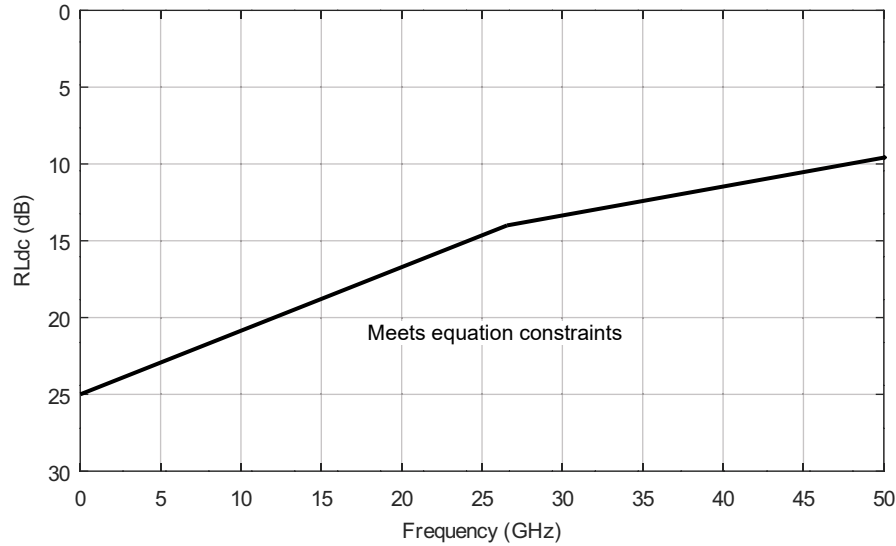


Figure 120G-5—Host output common-mode to differential-mode return loss

120G.3.1.2 Host output effective return loss (ERL)

The ERL of the host output at TP1a is computed using the procedure in 93A.5 with the values in Table 120G-2 and Table 120F-8, and with the value of T_{fx} equal to twice the delay between the test fixture test connector and the test fixture host-facing connection minus 0.2 ns.

Host output ERL at TP1a shall be greater than or equal to ERL (min) specified in Table 120G-1.

Table 120G-2—Host output and input ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.01	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	800	UI
Equalizer length associated with reflection signal	N_{bx}	0	UI
Tukey window flag	tw	1	—

120G.3.1.3 Differential termination mismatch

Differential termination mismatch is defined in 86A.5.3.2.

120G.3.1.4 Transition time

Transition time is defined in 120E.3.1.5, with the exception that the waveform is observed through a fourth-order Bessel-Thomson low-pass response with a 3 dB bandwidth of 40 GHz.

120G.3.1.5 Host output eye height and vertical eye closure (VEC)

Figure 120G–6 depicts an example host output eye height and vertical eye closure (VEC) test configuration. Host output eye height and VEC are measured at TP1a using compliance boards defined in 120G.5.4. Eye height and VEC are measured according to the method described in 120G.5.2.

All counter-propagating signals are asynchronous to the co-propagating signals using the PRBS13Q²¹ (see 120.5.11.2.1) or PRBS31Q²² (see 120.5.11.2.2) pattern, or a valid 100GBASE-R, 200GBASE-R, or 400GBASE-R signal. For the case where PRBS13Q or PRBS31Q are used with a common clock, there is at least 31 UI delay between the patterns on one lane and any other lane, so that the symbols on each lane are not correlated. The crosstalk generator is calibrated at TP4 using a test system with a response as defined in 120G.3.1, rather than the reference receiver of 120G.5.2, with target differential peak-to-peak voltage of 845 mV and transition time (see 120G.3.1.4) of 8.5 ps.

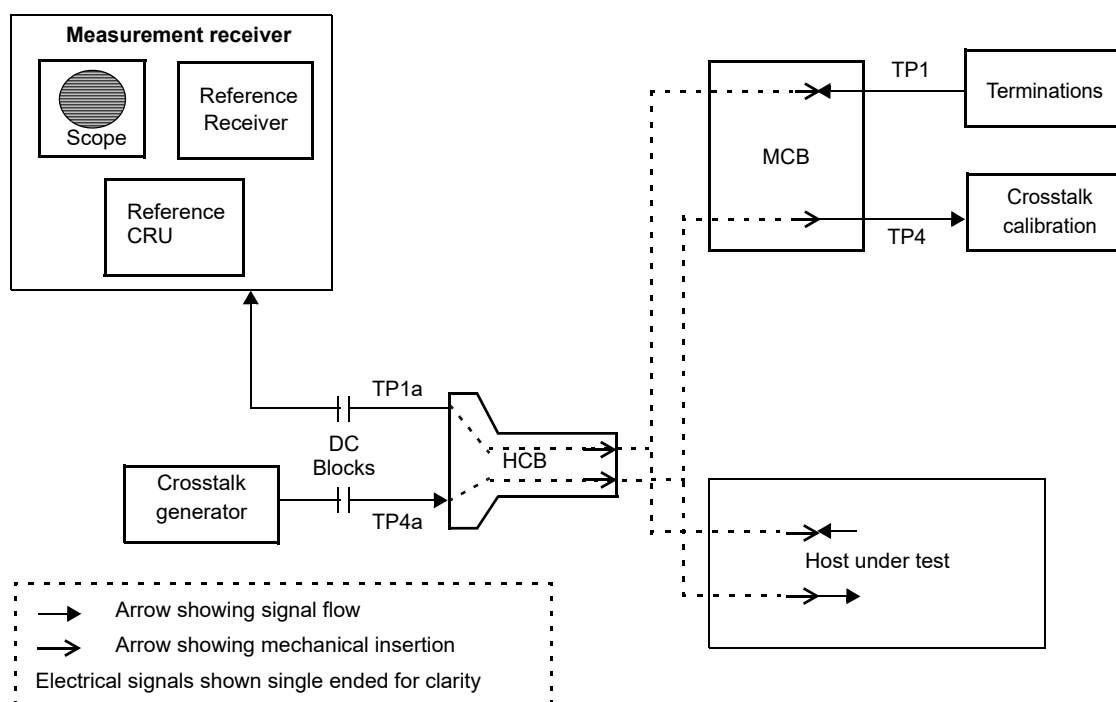


Figure 120G-6—Example host output test configuration

120G.3.2 Module output characteristics

The module output shall meet the specifications given in Table 120G–3. The low-frequency 3 dB cutoff of the output AC-coupling within the module should be less than 100 kHz. A test system with a fourth-order Bessel-Thomson low-pass response with 40 GHz 3 dB bandwidth is to be used for all output signal measurements, unless otherwise specified.

²¹ PRBS13O is also referred to as Pattern 4 for some PAM4 optical PMDs (e.g., see [Table 121–9](#)).

²² PRBS31Q is also referred to as Pattern 3 for some PAM4 optical PMDs (e.g., see [Table 121-9](#)).

Table 120G–3—Module output characteristics at TP4

Parameter	Reference	Value	Units
Signaling rate, each lane (nominal)		53.125 ^a	GBd
Peak-to-peak AC common-mode voltage (max) Low-frequency, $V_{CM_{LF}}$ Full-band, $V_{CM_{FB}}$	120G.5.1	32 80	mV
Differential peak-to-peak output voltage (max) Short mode Long mode	120G.5.1	600 845	mV mV
Eye height (min)	120G.3.2.2	15	mV
Vertical eye closure, VEC (max)	120G.3.2.2	12	dB
Common-mode to differential-mode return loss, RL_{dc} (min)	120G.3.1.1	Equation (120G–1)	dB
Effective return loss, ERL (min)	120G.3.2.3	8.5	dB
Differential termination mismatch (max)	120G.3.1.3	10	%
Transition time (min)	120G.3.1.4	8.5	ps
DC common-mode voltage tolerance (range) Upper limit Lower limit	120G.3.2.4	2.85 -0.35	V V

^a The signaling rate range is derived from the PMD receiver input.

120G.3.2.1 Module output modes

The module output shall support two modes: short and long. The means of controlling the module output mode is implementation dependent. For each output mode, the module shall meet the requirements for eye height (min) and VEC (max) in Table 120G–3 for both near-end and far-end measurements (see 120G.3.2.2.1).

One method for configuring the module output mode is to configure the module for a particular host electrical interface. The mapping of the combination of IEEE 802.3 interface type and module output mode to a host electrical interface may be determined from Table 120G–4.

Table 120G–4—Module output mode mapping

IEEE 802.3 interface type	Module output mode	Host electrical interface
100GAUI-1 C2M	short	100GAUI-1-S C2M
100GAUI-1 C2M	long	100GAUI-1-L C2M
200GAUI-2 C2M	short	200GAUI-2-S C2M
200GAUI-2 C2M	long	200GAUI-2-L C2M
400GAUI-4 C2M	short	400GAUI-4-S C2M
400GAUI-4 C2M	long	400GAUI-4-L C2M

120G.3.2.2 Module output eye height and VEC

Figure 120G–7 depicts an example module output eye height and VEC test configuration. Module output eye height and VEC are measured at TP4 using compliance boards defined in 120G.5.4. For each module output mode, eye height and VEC are measured according to the method described in 120G.3.2.2.1 using both the near-end and far-end host channels.

NOTE—Driving the module output using an internally generated clock, rather than the recovered clock from the PMD input, might create output signals that are more optimistic than full-link operation.

All counter-propagating signals are asynchronous to the co-propagating signals using the PRBS13Q²³ (see 120.5.11.2.1) or PRBS31Q²⁴ (120.5.11.2.2) pattern, or a valid 100GBASE-R, 200GBASE-R, or 400GBASE-R signal. For the case where PRBS13Q or PRBS31Q are used with a common clock, there is at least 31 UI delay between the patterns on one lane and any other lane, so that the symbols on each lane are not correlated. The crosstalk generator is calibrated at TP1a using a test system with a response as defined in 120G.3.1, rather than the reference receiver of 120G.5.2, with target differential peak-to-peak voltage of 750 mV and transition time (see 120G.3.1.4) of 10 ps for short mode and 15 ps for long mode.

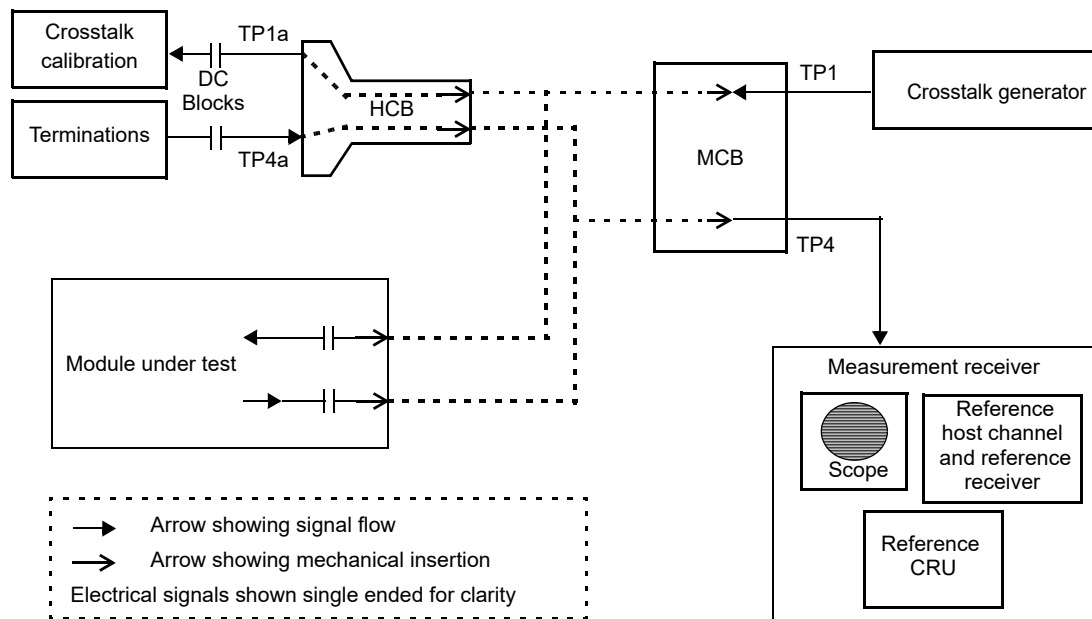


Figure 120G–7—Example module output test configuration

²³ PRBS13Q is also referred to as Pattern 4 for some PAM4 optical PMDs (e.g., see Table 121–9).

²⁴ PRBS31Q is also referred to as Pattern 3 for some PAM4 optical PMDs (e.g., see Table 121–9).

120G.3.2.2.1 Near-end and far-end eye measurement methodology

The signal measured at TP4 is first convolved with a reference host channel. The reference host channel is the host receiver printed circuit board (PCB) signal path $S^{(\text{HOSPR})}$ defined in 162.11.7.1.1 with the exceptions that the length z_p for each test is provided in Table 120G–5, and C_0 and C_1 are both 0 nF. The eye height and VEC are measured using the method in 120G.5.2.

Table 120G–5—PCB length for module output measurements

Module output mode	Host channel type	PCB length, z_p (mm)
Short	near-end	0
Short	far-end	133
Long	near-end	80
Long	far-end	244.7

120G.3.2.3 Module output ERL

The ERL of the module output at TP4 is computed using the procedure in 93A.5 with the values in Table 120G–6 and Table 120F–8, and with the value of T_{fx} equal to twice the delay between the test fixture test connector and the test fixture module-facing connection minus 0.2 ns.

Module output ERL at TP4 shall be greater than or equal to ERL (min) specified in Table 120G–3.

Table 120G–6—Module output and input ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.01	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	400	UI
Equalizer length associated with reflection signal	N_{bx}	0	UI
Tukey window flag	tw	1	—

120G.3.2.4 Module output common-mode voltage tolerance

DC common-mode voltage is generated by the host. A compliant module meets the output specifications with any DC common-mode voltage (see 120G.5.1), as measured at TP4, within the range specified in Table 120G–3.

NOTE—The specified voltages allow for the effects of ground offset voltage.

120G.3.3 Host input characteristics

The host input shall meet the specifications given in Table 120G–7. Channel equalization is provided by an adaptive equalizer in the host.

Table 120G–7—Host input characteristics

Parameter	Reference	Test point	Value	Units
Signaling rate, each lane (range)	120G.3.3.1	TP4a	53.125 ± 100 ppm	GBd
Differential peak-to-peak input voltage tolerance (min) for short mode for long mode	120G.5.1	TP4	600 845	mV
Peak-to-peak AC common-mode voltage tolerance (min) Low-frequency, $V_{CM_{LF}}$ Full-band, $V_{CM_{FB}}$	120G.5.1	TP4	32 80	mV
Differential-mode to common-mode return loss, RL_{cd} (min)	120G.3.3.3	TP4a	Equation (120G–2)	dB
Effective return loss, ERL (min)	120G.3.3.4	TP4a	7.3	dB
Host stressed input tolerance	120G.3.3.5	TP4	See 120G.3.3.5	
Differential termination mismatch (max)	120G.3.1.3	TP4a	10	%
Common-mode voltage ^a Min Max	120G.5.1	TP4a	–0.3 2.8	V

^a Generated by host, referred to host ground.

120G.3.3.1 Host input signaling rate

The host input shall comply with the requirements of 120G.3.3.5 for any signaling rate in the range specified in Table 120G–7.

120G.3.3.2 Host input AC common-mode voltage tolerance

A host input shall meet all other specifications with both $V_{CM_{LF}}$ and $V_{CM_{FB}}$ (see 120G.5.1) up to the limits specified in Table 120G–7.

120G.3.3.3 Host input differential-mode to common-mode return loss

The host input differential-mode to common-mode return loss shall meet Equation (120G–2) as illustrated in Figure 120G–8.

$$RL_{cd}(f) \geq \begin{cases} 22 - 16(f/53.125) & 0.05 \leq f \leq 26.56 \\ 19 - 10(f/53.125) & 26.56 < f \leq 50 \end{cases} \quad (120G-2)$$

where

RL_{cd} is the differential-mode to common-mode return loss in dB
 f is the frequency in GHz

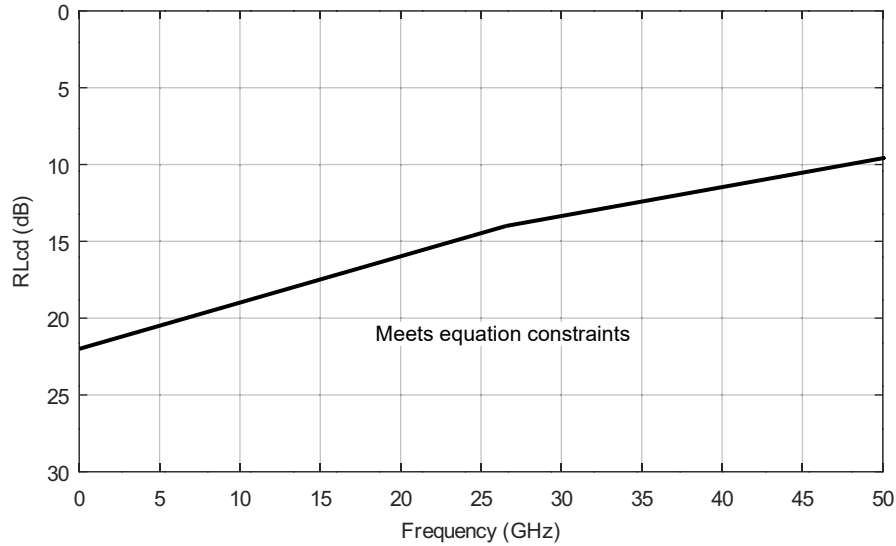


Figure 120G-8—Host input differential-mode to common-mode return loss

120G.3.3.4 Host input ERL

The ERL of the host input at TP4a is computed using the procedure in 93A.5 with the values in Table 120G-2 and Table 120F-8.

Host input ERL at TP4a shall be greater than or equal to ERL (min) specified in Table 120G-7.

120G.3.3.5 Host stressed input tolerance

Host stressed input tolerance is defined by the procedure described in 120G.3.3.5.1 through 120G.3.3.5.3.

The host under test shall meet the BER requirement in 120G.1.1:

- For either the short or long mode.
- With all sinusoidal jitter cases in Table 162-17.
- For any signaling rate in the range given in Table 120G-7.

120G.3.3.5.1 Host stressed input test setup

The host stressed input test setup is illustrated in Figure 120G-9. The stressed signal is applied at TP4a and is calibrated at TP4.

The stressed signal includes the following impairments:

- Sinusoidal jitter, random jitter, and bounded uncorrelated jitter.
- Counter-propagating crosstalk signals.

The pattern generator output equalization functional behavior is equivalent to the model shown in Figure 120F–3. The tap coefficients are not specified with the exception that the “no equalization” state is defined as setting coefficient $c(0)$ to 1 and all other coefficients to 0.

Bounded uncorrelated jitter may not be available in all stressed pattern generators or bit error ratio testers. It can be generated by driving the pattern generator external jitter modulation input with a low-pass filtered pseudo-random pattern. The pattern should be either PRBS7 or PRBS9 (see 83.5.10) with a signaling rate approximately 1/20 of the stressed pattern signaling rate (e.g., 2.656 GBd). The clock source for the PRBS generator is asynchronous to the pattern generator clock source. The low-pass filter has 20 dB/decade roll-off with a –3 dB corner frequency between 150 MHz and 300 MHz.

A crosstalk generator for counter-propagating signals representing the host output is used for calibration of the stress. The counter-propagating signals are asynchronous to the pattern generator. This crosstalk generator is not present in the test, being replaced by the output of the host under test.

The measurement receiver used for test calibration includes:

- A clock recovery unit (CRU) that acts as a high-pass jitter filter with a 3 dB corner frequency of 4 MHz and a slope of 20 dB/decade.
- A reference host channel to be configured for short or long mode far-end test calibration or long-mode near end verification as specified in 120G.3.2.2.1.
- A reference receiver as specified in 120G.5.2.

The test system meets the ERL specification given in 120G.3.2.3 when measured at TP4.

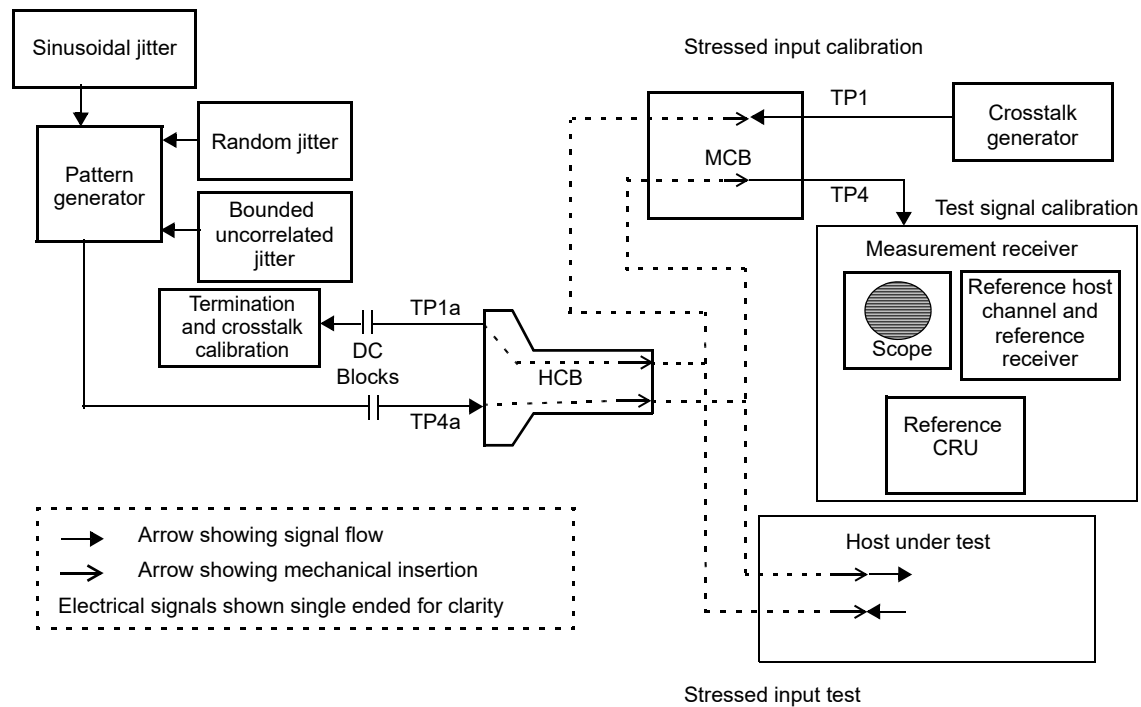


Figure 120G-9—Example host stressed input test

120G.3.3.5.2 Host stressed input test calibration

The host stressed input signal is calibrated by the following procedure.

- The pattern generator is set to generate a PRBS13Q pattern (see 120.5.11.2.1). The transition time (see 120G.3.1.4) measured at TP4a, with the pattern generator output equalization configured for “no equalization”, is as specified in Table 120G-8. The initial signal level is set to the differential peak-to-peak input voltage tolerance given in Table 120G-7.
- Sinusoidal jitter is applied with frequency and amplitude per case F in Table 162-17.
- Random jitter and bounded uncorrelated jitter are added to the sinusoidal jitter such that the jitter profile of the signal at TP4a approximates J_{RMS} (max) and J_{4u} (max) and complies with the even-odd jitter (max) specification in Table 120F-1. These are initial jitter values, that will be modified by the addition of crosstalk in step e) and adjustment of random jitter in step g).
- The HCB is plugged into the MCB.
- The counter-propagating crosstalk signals are calibrated at TP1a using a test system with a response as defined in 120G.3.1, rather than the reference receiver of 120G.5.2, to the differential peak-to-peak voltage and transition time (see 120G.3.1.4) specified in Table 120G-8. The crosstalk signal transition time and amplitude are calibrated with a PRBS13Q pattern. The crosstalk pattern is changed to PRBS31Q (see 120.5.11.2.2), scrambled idle (see 82.2.11 and 119.2.4.9), or another valid 100GBASE-R, 200GBASE-R, or 400GBASE-R signal for stressed signal calibration (see step g)). If the PRBS31Q pattern is used with a common clock, there is at least 31 UI delay between the PRBS31Q patterns on one lane and any other lane.
- The reference host channel is configured in the same way as in 120G.3.2.2.1 using the parameters in Table 120G-5 for host channel type and the requested module mode (short or long).

- g) Eye height and VEC are measured at TP4 as described in 120G.5.2. The pattern generator amplitude and random jitter are adjusted, while the pattern generator output equalization and reference receiver settings are adjusted to minimize far-end VEC, so that the far-end eye height of the smallest eye matches the target value, near-end VEC is within the limits in Table 120G–8, and differential peak-to-peak output voltage, near-end VEC, and eye height are within the limits in Table 120G–3. The differential peak-to-peak voltage measured at TP4 does not exceed the differential peak-to-peak input voltage tolerance given in Table 120G–7.

Table 120G–8—Host stressed input parameters

Parameter	Value	Units
Pattern generator transition time (target)	9	ps
Applied peak-to-peak sinusoidal jitter	Table 162–17	—
Eye height (target)	15	mV
Vertical eye closure, VEC (min)	12	dB
Vertical eye closure, VEC (max)	12.5	dB
Crosstalk differential peak-to-peak voltage	750	mV
Crosstalk transition time for short mode for long mode	10 15	ps

120G.3.3.5.3 Host stressed input test procedure

After the stress has been calibrated, the pattern generator is set to generate PRBS31Q, scrambled idle, or another valid 100GBASE-R, 200GBASE-R, or 400GBASE-R sequence. The HCB is unplugged from the MCB and is plugged into the host under test. The host electrical output is enabled on all lanes with any of the patterns above. The test is repeated with sinusoidal jitter set to each of the six cases in Table 162–17.

If the test is performed with PRBS31Q, the BER for the AUI under test may be calculated using the bit error counter in the PMA test pattern checker (see 120.5.11.2.2) as the number of bit errors divided by the number of received bits. The BER for the AUI under test is the average of the BER of each of its lanes.

If the test is performed with scrambled idle or another valid 100GBASE-R, 200GBASE-R, or 400GBASE-R sequence, the BER for the AUI under test may be calculated using the FEC decoder error counters (see 91.6 and 119.3.1), as the number of FEC symbol errors divided by the number of received bits.

The number of received bits may be estimated based on the test time.

Methods of extracting the received bit pattern and counting errors other than the ones described above may be used if they generate equivalent results.

120G.3.4 Module input characteristics

The module input shall meet the specifications given in Table 120G–9. Channel equalization is provided by an adaptive equalizer in the module.

Table 120G–9—Module input characteristics

Parameter	Reference	Test point	Value	Units
Signaling rate, each lane (range)	120G.3.4.1	TP1	53.125 ± 100 ppm	GBd
Differential pk-pk voltage tolerance (min)	120G.5.1	TP1a	750	mV
Peak-to-peak AC common-mode voltage tolerance (min) Low-frequency, $V_{CM_{LF}}$ Full-band, $V_{CM_{FB}}$	120G.5.1	TP1a	32 80	mV
Differential-mode to common-mode return loss, RL_{cd} (min)	120G.3.3.3	TP1	Equation (120G–2)	dB
Effective return loss, ERL (min)	120G.3.4.4	TP1	8.5	dB
Differential termination mismatch (max)	120G.3.1.3	TP1	10	%
Module stressed input tolerance	120G.3.4.3	TP1a	See 120G.3.4.3	
Single-ended voltage tolerance range (min)	120G.5.1	TP1a	–0.4 to 3.3	V
DC common-mode voltage tolerance (range) Upper limit Lower limit	120G.3.4.5	TP1	2.85 –0.35	V V

120G.3.4.1 Module input signaling rate

The module input shall comply with the requirements of 120G.3.4.3 for any signaling rate in the range specified in Table 120G–9.

120G.3.4.2 Module input AC common-mode voltage tolerance

A module input shall meet all other specifications with both $V_{CM_{LF}}$ and $V_{CM_{FB}}$ (see 120G.5.1) up to the limits specified in Table 120G–9.

120G.3.4.3 Module stressed input tolerance

Module stressed input tolerance is defined by the procedure described in 120G.3.4.3.1 through 120G.3.4.3.3.

The module under test shall meet the BER requirement in 120G.1.1:

- For both the high-loss and low-loss cases (see 120G.3.4.3.2), calibrated separately.
- With all sinusoidal jitter cases in Table 162–17.
- For any signaling rate in the range given in Table 120G–9.

120G.3.4.3.1 Module stressed input test setup

The module stressed input test setup is illustrated in Figure 120G–10. The stressed signal is applied at TP1 and is calibrated at TP1a.

The stressed signal includes the following impairments:

- Sinusoidal jitter, random jitter, and bounded uncorrelated jitter.
- Frequency-dependent attenuation representing the host channel, which may be implemented with PCB traces. The frequency-dependent attenuation is used only for the high-loss case (see 120G.3.4.3.2).
- Counter-propagating crosstalk signals.

The pattern generator output equalization is described in 120G.3.3.5.1.

Bounded uncorrelated jitter may not be available in all stressed pattern generators or bit error ratio testers. It can be generated as described in 120G.3.3.5.2.

A crosstalk generator for counter-propagating signals representing the module output is used for calibration of the stress. The counter-propagating signals are asynchronous to the pattern generator. This crosstalk generator is not present in the test, being replaced by the output of the module under test.

The measurement receiver used for test calibration includes:

- A clock recovery unit (CRU) that acts as a high-pass jitter filter with a 3 dB corner frequency of 4 MHz and a slope of 20 dB/decade.
- A reference receiver as specified in 120G.5.2.

The test system meets the ERL specification given in 120G.3.1.2 when measured at TP1a.

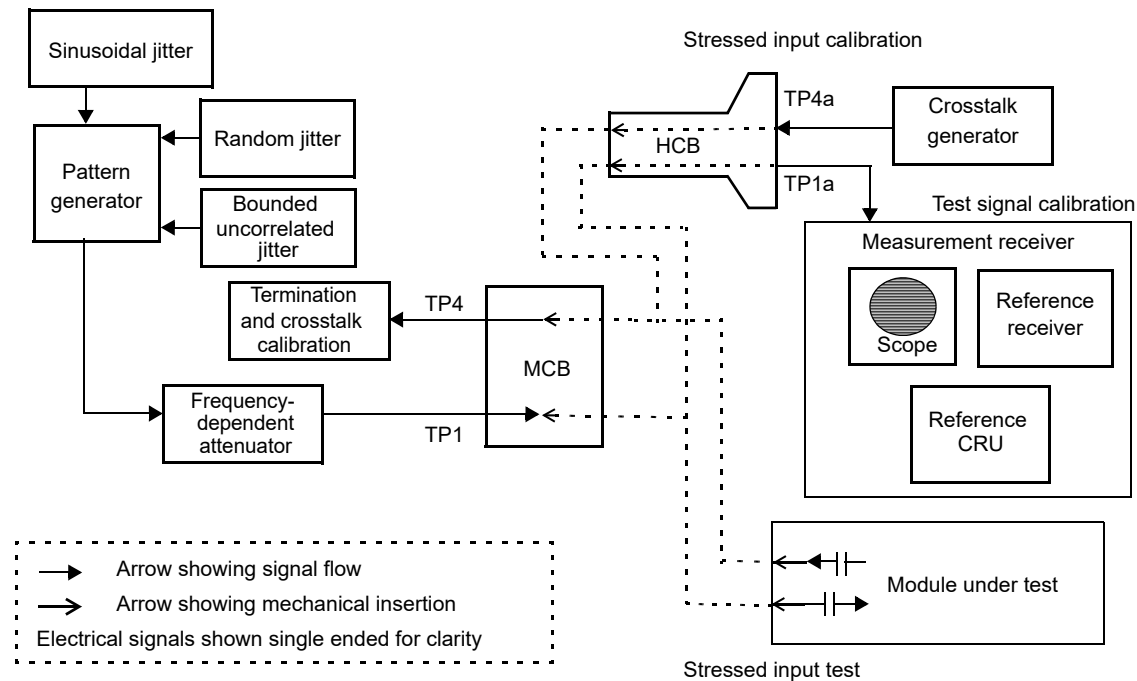


Figure 120G-10—Example module stressed input test

120G.3.4.3.2 Module stressed input test calibration

The stressed input signal is calibrated by the following procedure.

- The pattern generator is set to generate a PRBS13Q pattern (see 120.5.11.2.1). The transition time (see 120G.3.1.4) measured at the output of the pattern generator, with the pattern generator output equalization configured for “no equalization” (see 120G.3.3.5.1), is as specified in Table 120G-10. The initial signal level is set to the differential peak-to-peak input voltage tolerance given in Table 120G-9.
- Sinusoidal jitter is applied with frequency and amplitude per case F in Table 162-17.
- Random jitter and bounded uncorrelated jitter are added to the sinusoidal jitter such that the jitter profile of the signal at the output of the pattern generator approximates $J_{RMS}(\max)$ and $J_{4u}(\max)$ and complies with the even-odd jitter (max) specification in Table 120F-1. These are initial jitter values, that will be modified by the addition of crosstalk in step e) and adjustment of random jitter in step g).
- The HCB is plugged into the MCB.
- The counter-propagating crosstalk signals are calibrated at TP4 using a test system with a response as defined in 120G.3.1, rather than the reference receiver of 120G.5.2, to the differential peak-to-peak voltage and transition time (see 120G.3.1.4) specified in Table 120G-10. The crosstalk signal transition time and amplitude are calibrated with a PRBS13Q pattern. The crosstalk pattern is changed to PRBS31Q (see 120.5.11.2.2), scrambled idle (see 82.2.11, 119.2.4.9), or another valid 100GBASE-R, 200GBASE-R, or 400GBASE-R signal for stressed signal calibration (see step g)). If the PRBS31Q pattern is used with a common clock, there is at least 31 UI delay between the PRBS31Q patterns on one lane and any other lane.
- For the low-loss signal calibration, the output of the pattern generator is fed directly to the MCB input (TP1). For the high-loss signal calibration, the frequency-dependent attenuator is configured such that the insertion loss of the test channel from the output of the pattern generator to TP1a

approximates the target insertion loss, $ILDD_{channel}$, given by Equation (120G–3). The target insertion loss is 18.2 dB at 26.56 GHz, representing 16 dB channel loss with an additional 2.2 dB, which provides some allowance for the difference between the effective insertion loss of the pattern generator output and the worst-case host transmitter package loss. To achieve the target test channel insertion loss, the scattering parameters of the frequency-dependent attenuator should approximate those for a PCB transmission line calculated from Equation (93A–13) and Equation (93A–14) using $z_p = 296$ mm in length and the relevant parameter values given in Table 162–21. Additional frequency-independent delay may be included in the test channel. The magnitude of the insertion loss of the target frequency-dependent attenuator scattering parameters, $ILDD_{FDA}$, is represented by Equation (120G–4). $ILDD_{channel}$ and $ILDD_{FDA}$ are illustrated in Figure 120G–11.

- g) Eye height and VEC are measured at TP1a as described in 120G.5.2. For the high-loss case, an exception is made that the reference receiver CTLE is limited to settings where $g_{DC} + g_{DC2}$ is less than or equal to -10.5 dB.
- h) The pattern generator amplitude, output equalization, and random jitter are adjusted together to minimize VEC, so that the eye height of the smallest eye matches the target value, VEC is within the limits in Table 120G–10, and the differential peak-to-peak voltage measured at TP1a does not exceed the differential peak-to-peak input voltage tolerance given in Table 120G–9. The pattern generator output equalization has to be set such that for the resulting signal, the same VEC is achieved with or without the limitations on g_{DC} and g_{DC2} in item g).

Other calibration procedures resulting in a signal that meets these requirements may be used.

Table 120G–10—Module stressed input parameters

Parameter	Value	Value
Pattern generator transition time (target)	9	ps
Applied peak-to-peak sinusoidal jitter	Table 162–17	—
Eye height (target)	10	mV
Vertical eye closure, VEC (min)	12	dB
Vertical eye closure, VEC (max)	12.5	dB
Crosstalk differential peak-to-peak voltage	845	mV
Crosstalk transition time	8.5	ps

$$ILdd_{channel}(f) = 1.7962\sqrt{f} + 0.2463f + 0.003405f^2 \quad (120G-3)$$

where

$ILdd_{channel}(f)$ is the target pattern generator output to TP1a insertion loss in dB
 f is the frequency in GHz

$$ILdd_{FDA}(f) = 0.981 \sqrt{f} + 0.2463f \quad (120G-4)$$

where

$ILdd_{FDA}(f)$ is the target high-loss frequency-dependent attenuator insertion loss in dB
 f is the frequency in GHz

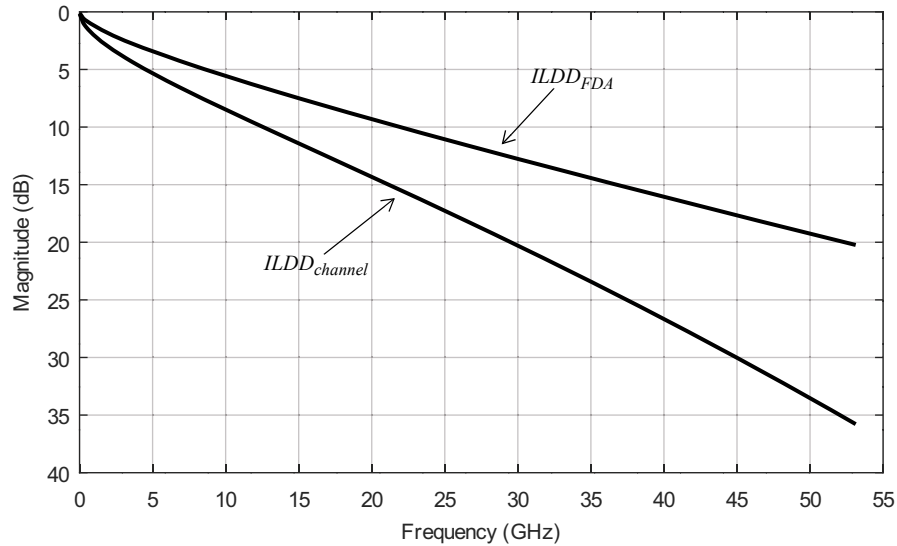


Figure 120G-11—Module stressed input insertion loss targets

120G.3.4.3.3 Module stressed input test procedure

After the stress has been calibrated, the pattern generator is set to generate PRBS31Q, scrambled idle, or another valid 100GBASE-R, 200GBASE-R, or 400GBASE-R sequence. The HCB is unplugged from the MCB and the module under test is plugged into the MCB. The module electrical output is enabled on all lanes with any of the patterns above. The test is repeated with sinusoidal jitter set to each of the six cases in Table 162-17.

If the test is performed with PRBS31Q, the BER for the AUI under test may be calculated using the bit error counter in the PMA test pattern checker (see 120.5.11.2.2) as the number of bit errors divided by the number of received bits. The BER for the AUI under test is the average of the BER of each of its lanes.

If the test is performed with scrambled idle or another valid 100GBASE-R, 200GBASE-R, or 400GBASE-R sequence, the BER for the AUI under test may be calculated by placing the module under test into local loopback (see 120.5.9) and feeding the module output into a compliant host or its equivalent. The BER for the AUI under test is calculated using the FEC decoder error counters (see 91.6 and 119.3.1), as the number of FEC symbol errors divided by the number of received bits.

The number of received bits may be estimated based on the test time.

Methods of extracting the received bit pattern and counting errors other than the ones described above may be used if they generate equivalent results.

120G.3.4.4 Module input ERL

The ERL of the module input at TP1 is computed using the procedure in 93A.5 with the values in Table 120G–6 and Table 120F–8.

Module input ERL at TP1 shall be greater than or equal to ERL (min) specified in Table 120G–9.

120G.3.4.5 Module input common-mode voltage tolerance

DC common-mode voltage is generated by the host. A compliant module meets the input specifications with any DC common-mode voltage (see 120G.5.1), as measured at TP1, within the range specified in Table 120G–9.

NOTE—The specified voltages allow for the effects of ground offset voltage.

120G.4 Channel insertion loss (recommended)

For worst-case package loss in the host, the insertion loss of the channel between the host and module C2M components (see Figure 120G–2) is recommended to meet Equation (120G–5), which is illustrated in Figure 120G–12. Note that for this equation the insertion loss at the Nyquist frequency is less than or equal to 16 dB. The insertion loss budget is illustrated in Figure 120G–2.

$$IL_{dd}(f) \leq \begin{cases} 0.05 + 1.8\sqrt{f} + 0.2513f & 0.01 \leq f \leq 26.56 \\ -12.4181 + 1.07f & 26.56 < f \leq 40 \end{cases} \quad (120G-5)$$

where

$IL_{dd}(f)$ is the channel insertion loss in dB
 f is the frequency in GHz

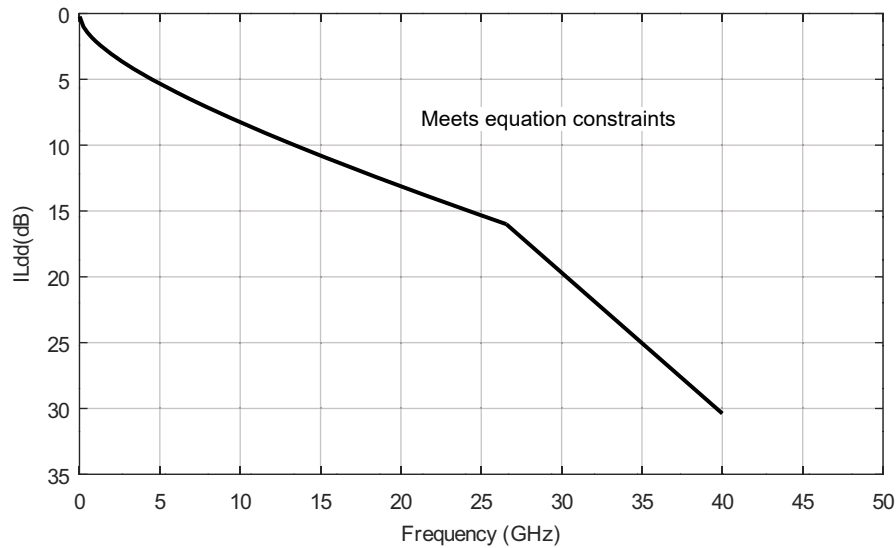


Figure 120G–12—Recommended channel insertion loss

120G.5 Measurement methodology

This subclause describes common measurement tools and methodology.

120G.5.1 Signal levels

The signal levels are as defined in 120E.3.1.2.

Low-frequency and full-band peak-to-peak AC common-mode voltage, $V_{CM_{LF}}$ and $V_{CM_{FB}}$, respectively, are defined by the method specified in 162.9.4.4 with the following exceptions:

- The peak-to-peak AC common-mode voltage is defined as the AC common-mode voltage range measured at TP1a or TP4 that includes all but 10^{-5} of the measured distribution, from 0.000005 to 0.999995 of the cumulative distribution.
- The condition for transmitter equalization to be turned off does not apply.

120G.5.2 Eye opening measurement method

The eye opening parameters eye height and VEC are measured with the effect of a reference receiver that includes receiver input referred noise, a continuous-time filter as defined in 93A.1.4.3, a receiver noise filter as defined in 93A.1.4.1, and a decision-feedback equalizer as defined in 93A.1.6, using the parameters specified in Table 120G–11. All parameters in Table 120G–11 apply to TP1a, TP4 far-end, and TP4 near-end unless indicated otherwise. The pattern used for output eye diagram measurements is PRBS13Q (see 120.5.11.2.1). Eye height and VEC parameters are defined by the following procedure.

Table 120G–11—Eye opening reference receiver parameter values

Parameter	Symbol	Value	Units
Signaling rate	f_b	53.125	GBd
Receiver 3 dB bandwidth	f_r	$0.75 \times f_b$	GHz
Continuous time filter, DC gain for TP1a Range for $g_{DC2} = 0$ Range for $-1 \leq g_{DC2} < 0$ Range for $-2 \leq g_{DC2} < -1$ Range for $-3 \leq g_{DC2} < -2$ Step size	g_{DC}	–2 to –9 –2 to –11 –4 to –10 –4 to –9 1.0	dB
Continuous time filter, DC gain 2 for TP1a Range Step size	g_{DC2}	–3 to 0 0.5	dB
Continuous time filter, DC gain for TP4 near-end Range Step size	g_{DC}	–5 to –1 1.0	dB
Continuous time filter, DC gain 2 for TP4 near-end Range Step size	g_{DC2}	–2 to 0 0.5	dB
Continuous time filter, DC gain for TP4 far-end Range Step size	g_{DC}	–9 to –2 1.0	dB

Table 120G–11—Eye opening reference receiver parameter values (*continued*)

Parameter	Symbol	Value	Units
Continuous time filter, DC gain 2 for TP4 far-end Range Step size	g_{DC2}	–3 to –1 0.5	dB
Continuous time filter, zero frequency for $g_{DC} = 0$	f_z	12.58	GHz
Continuous time filter, pole frequencies	f_{p1} f_{p2}	20 28	GHz GHz
Continuous time filter, low-frequency pole/zero	f_{LF}	$f_b / 40$	GHz
Decision feedback equalizer (DFE) length	N_b	4	UI
Normalized DFE coefficient maximum limit $n = 1$ $n = 2$ $n = 3$ or 4	$bb_{\max}(n)$	0.4 0.15 0.1	— — —
Normalized DFE coefficient minimum limit $n = 1$ $n = 2$ $n = 3$ or 4	$bb_{\min}(n)$	0.1 –0.15 –0.05	— — —
One-sided noise spectral density	η_0	4.1×10^{-8}	V ² /GHz

Perform the following step once:

- a) Capture the PRBS13Q signal $y_1(k)$ with the effect of low-pass response equivalent to the specified receiver noise filter with associated parameter f_r in Table 120G–11 (instead of the test system response specified in 120G.3.1), and using a clock recovery unit with a corner frequency of 4 MHz and slope of 20 dB/decade. The signal is captured such that samples, after any post-processing as necessary, are evenly distributed throughout the measurement window and are numerous enough for an accurate normalized cumulative distribution function (CDF) to a probability of 10^{-5} without extrapolation.

Perform the following steps for each valid combination of g_{DC} and g_{DC2} as specified in Table 120G–11:

- b) Compute the response $y_2(k)$ by applying the effect of the continuous time filter to $y_1(k)$ using the associated parameters in Table 120G–11.
- c) Compute the linear fit pulse response $p_2(k)$ using the method defined in 162.9.4.1.1 with parameter M the same as for step a), D_p equal to 3, and N_p equal to 200.
- d) Compute the DFE sampling phase t_s and tap weights $b(n)$ for $p_2(k)$ according to the methodology in 93A.1.6 using the associated parameters in Table 120G–11.
- e) Compute the receiver input signal $y_{rx}(k)$ by applying the effect of the DFE to $y_2(k)$ using the sampling phase t_s and tap weights $b(n)$ determined in the previous step.
- f) Compute the variance of the noise at the output of the receive equalizer σ_N^2 based on the one-sided spectral density η_0 , provided in Table 120G–11, referred to the receiver noise filter input per Equation (93A–35).
- g) Compute an eye diagram from $y_{rx}(k)$, including the effect of Gaussian noise with variance calculated in the previous step, but taking into account that some noise from the measurement instrument is already in $y_{rx}(k)$.
- h) Compute Vmid, Vupp, Vlow, VCmid, VCupp, VClow from the eye diagram using 120E.4.2 steps 4) through 6) with the exception that the CDF of the signal voltage is calculated from the set of voltage samples within the time interval $t_s \pm 0.05$ UI and with accumulated probability for each sample

weighted by the function $w(t)$ defined by Equation (120G–6). As an example, $UPPCDF1$ may be calculated using Equation (120G–8) from a set of N samples $\{v_i, t_i\}$.

- i) The eye height is the minimum of V_{mid} , V_{upp} , and V_{low} .
- j) Compute VEC using Equation (120E–4) and the values computed in step h) with the exception that $VM3$, $VM2$, $VM1$, and $VM0$ are the average values over the time interval $t_s \pm 0.05$ UI instead of “within 0.025 UI of time TC_{mid} ”.

The values of eye height and VEC are the values obtained with the combination of g_{DC} and g_{DC2} that produces the minimum value of VEC where eye height also complies with the specification for eye height (min) as specified for the interface.

$$w(t) = \begin{cases} \frac{1}{\sqrt{2\pi}\sigma_r} \exp\left(-\frac{1}{2}\left(\frac{x(t)}{\sigma_r}\right)^2\right) & |x(t)| \leq 0.05 \\ 0 & |x(t)| > 0.05 \end{cases} \quad (120G-6)$$

$$x(t) = (t - t_s)f_b \quad (120G-7)$$

$$UPPCDF1(v) = \frac{1}{A} \sum_{i=1}^N w(t_i) u(v_i, VC_{upp}, v) \quad (120G-8)$$

$$u(x, a, b) = \begin{cases} 1 & a \leq x \leq b \\ 0 & \text{otherwise} \end{cases} \quad (120G-9)$$

$$A = \sum_{i=1}^N w(t_i) \quad (120G-10)$$

where:

$w(t)$	is the weighting function
$x(t)$	is the deviation from the sampling time t_s in UI
σ_r	is the standard deviation in UI and is equal to 0.02 UI
f_b	is the signaling rate in Hz
t	is time in seconds
A	is a factor to normalize the weighting function over the sample set

120G.5.3 Steady-state voltage

The steady-state voltage v_f is defined as the sum of the linear fit pulse $p(1)$ through $p(M \times N_p)$ divided by M with the specific equalization used by the transmitter. N_v is set equal to N_p . The linear fit procedure for obtaining $p(k)$ and the values of M and N_p are defined in 162.9.4.1.1.

120G.5.4 HCB/MCB characteristics

HCB characteristics are described in 162B.2 where the HCB performs the equivalent function as the TP2 or TP3 test fixture. The MCB characteristics are described in 162B.3 where the MCB performs the equivalent

functionality as the cable assembly test fixture. The mated compliance board characteristics are described in 162B.4.

120G.6 Protocol implementation conformance statement (PICS) proforma for Annex 120G, Chip-to-module 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2M), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2M), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2M)²⁵

120G.6.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Annex 120G, Chip-to-module 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2M), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2M), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2M), shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in [Clause 21](#).

120G.6.2 Identification

120G.6.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

120G.6.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3ck-2022, Annex 120G, Chip-to-module 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2M), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2M), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2M)
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3ck-2022.)	

Date of Statement	
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²⁵Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this annex so that it can be used for its intended purpose and may further publish the completed PICS.

120G.6.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
NOL	Number of differential AC-coupled lanes	120G.1	One, two, or four independent data paths in each direction for 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2M, respectively	M	Yes []
BER	Meets BER requirement	120G.1.1	See 120G.1.1	M	Yes []
*PCSPKG	PMA in same package as PCS.			O	Yes [] No []

120G.6.4 PICS proforma tables for Chip-to-module 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2M), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2M), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2M)

120G.6.4.1 Host output

Item	Feature	Subclause	Value/Comment	Status	Support
TH1	Signaling rate, each lane, PMA in same package as PCS	120G.3.1	Per Table 120G–1	PCSPKG:M	Yes [] N/A []
TH2	Signaling rate, each lane, PMA not in same package as PCS	120G.3.1	Derived from PMA input lanes	!PCSPKG:M	Yes [] N/A []
TH3	DC common-mode output voltage	120G.3.1	Per Table 120G–1	M	Yes []
TH4	Single-ended output voltage	120G.3.1	Per Table 120G–1	M	Yes []
TH5	Low-frequency peak-to-peak AC common-mode voltage	120G.3.1	Per Table 120G–1	M	Yes []
TH6	Full-band peak-to-peak AC common-mode voltage	120G.3.1	Per Table 120G–1	M	Yes []
TH7	Differential peak-to-peak voltage, transmitter disabled	120G.3.1	Per Table 120G–1	M	Yes []
TH8	Differential peak-to-peak voltage, transmitter enabled	120G.3.1	Per Table 120G–1	M	Yes []
TH9	Steady-state voltage	120G.3.1	Per Table 120G–1	M	Yes []
TH10	Eye height	120G.3.1	Per Table 120G–1	M	Yes []
TH11	VEC	120G.3.1	Per Table 120G–1	M	Yes []
TH12	Common-mode to differential-mode return loss	120G.3.1.1	Per Table 120G–1	M	Yes []
TH13	ERL	120G.3.1.2	Per Table 120G–1	M	Yes []
TH14	Differential termination mismatch	120G.3.1.3	Per Table 120G–1	M	Yes []
TH15	Transition time	120G.3.1.4	Per Table 120G–1	M	Yes []

120G.6.4.2 Module output

Item	Feature	Subclause	Value/Comment	Status	Support
TM1	Signal rate, each lane, nominal	120G.3.2	Per Table 120G–3	M	Yes []
TM2	Signal rate, each lane, range	120G.3.2	Derived from PMD receiver input.	M	Yes []
TM3	Low-frequency peak-to-peak AC common-mode voltage	120G.3.2	Per Table 120G–3	M	Yes []
TM4	Full-band peak-to-peak AC common-mode voltage	120G.3.2	Per Table 120G–3	M	Yes []
TM5	Peak-to-peak differential output voltage	120G.3.2	Per Table 120G–3	M	Yes []
TM6	Near-end eye height	120G.3.2	Per Table 120G–3	M	Yes []
TM7	Near-end VEC	120G.3.2	Per Table 120G–3	M	Yes []
TM8	Far-end eye height	120G.3.2	Per Table 120G–3	M	Yes []
TM9	Far-end VEC	120G.3.2	Per Table 120G–3	M	Yes []
TM10	Common-mode to differential-mode return loss	120G.3.2	Per Table 120G–3	M	Yes []
TM11	ERL	120G.3.2	Per Table 120G–3	M	Yes []
TM12	Differential termination mismatch	120G.3.2	Per Table 120G–3	M	Yes []
TM13	Transition time	120G.3.2	Per Table 120G–3	M	Yes []
TM14	DC common-mode voltage	120G.3.2	Per Table 120G–3	M	Yes []

120G.6.4.3 Host input

Item	Feature	Subclause	Value/Comment	Status	Support
RH1	Adaptive equalizer in host	120G.3.3		M	Yes []
RH2	Signaling rate tolerance, each lane	120G.3.3	Per Table 120G–7	M	Yes []
RH3	Differential peak-to-peak input voltage tolerance	120G.3.3	Per Table 120G–7	M	Yes []
RH4	Common-mode to differential-mode return loss	120G.3.3	Per Table 120G–7	M	Yes []
RH5	ERL	120G.3.3	Per Table 120G–7	M	Yes []
RH6	Differential termination mismatch	120G.3.3	Per Table 120G–7	M	Yes []
RH7	Common-mode voltage	120G.3.3	Per Table 120G–7	M	Yes []
RH8	Stressed input test	120G.3.3.5		M	Yes []

120G.6.4.4 Module input

Item	Feature	Subclause	Value/Comment	Status	Support
RH1	Adaptive equalizer in module	120G.3.4		M	Yes []
RH2	Signaling rate tolerance, each lane	120G.3.4	Per Table 120G–9	M	Yes []
RH3	Differential peak-to-peak input voltage tolerance	120G.3.4	Per Table 120G–9	M	Yes []
RH4	Common-mode to differential-mode return loss	120G.3.4	Per Table 120G–9	M	Yes []
RH5	ERL	120G.3.4	Per Table 120G–9	M	Yes []
RH6	Differential termination mismatch	120G.3.4	Per Table 120G–9	M	Yes []
RH7	Single-ended voltage tolerance	120G.3.4	Per Table 120G–9		
RH8	Common-mode voltage tolerance	120G.3.4	Per Table 120G–9	M	Yes []
RH9	Stressed input test	120G.3.4.3		M	Yes []